

 What is MPLADS-SATARK?

MPLADS-SATARK is an AI-powered forensic decision-support platform designed for District Magistrates (DMs), Vigilance Officers, and CAG Auditors. It automatically scans government civil proposals against **100% real MoSPI records** using **7 specialized sentinel engines** to catch corruption, overpricing, duplicate works, and ghost assets *before* public funds are disbursed.



Feature 1: VIDHI-KAVACH (Negative List & March Rush Shield)

STATUTORY
COMPLIANCE SENTRY

In Simple Words: An automatic rule checker that ensures MP funds are spent strictly according to official government laws and never on banned projects or year-end money rushes.

HOW IT WORKS:

- **Negative List Detection:** Scans project titles against **Annexure-I of the MPLADS 2023 Guidelines**, which strictly prohibits using public funds for places of worship (mandir/masjid/church), private commercial trusts, or clubhouses.
- **March Rush Detection:** Checks sanction dates against **General Financial Rules (GFR 2017 Rule 62)** to flag projects rushed in the last week of March just to exhaust unspent annual budget allocations.

Data Used: 176,925 MoSPI Works

Rulebook: GFR 2017 Rule 62 & MPLADS 2023 Annexure-I

 **Real Result:** Flagged 18,851 statutory breaches nationwide (15,953 Negative List breaches and 3,204 March Rush violations).



Feature 2: PUNAR-DRISHTI (NLP Twin Work & Duplicate Sentry)

NLP DUPLICATE
SENTRY

In Simple Words: A smart duplicate detector that prevents contractors and agencies from getting paid twice for the exact same civil work under slightly tweaked names.

HOW IT WORKS:

- Uses **Natural Language Processing (Jaccard + Levenshtein fuzzy string matching)** to compare new proposal descriptions against all existing works in the same constituency.
- Even if a title is rephrased (e.g. "PCC Road near Village Well" vs "Construction of PCC Road near Village Well"), PUNAR-DRISHTI calculates a 95%+ text similarity match and flags the clone.

Data Used: Real MoSPI Work Descriptions

Tech: NLP Jaccard + Levenshtein Engine

Real Result: Caught 10,204 duplicate work claims across India (6,965 exact 100% clone works).



Feature 3: ARTHA-DARPAN (CPWD Rates & Cost Sentry)

COST INFLATION
SENTRY

In Simple Words: A price monitor that makes sure government construction projects are not overpriced or artificially inflated compared to standard market rates.

HOW IT WORKS:

- Compares proposed sanction costs against the official **CPWD Delhi Schedule of Rates (DSR 2023-24)** across 108 State-Category benchmarks.
- If a standard rural school room benchmark is ₹4 Lakhs but a proposal requests ₹12 Lakhs, ARTHA-DARPAN calculates a **+200% cost deviation anomaly** and alerts the District Collector.

Data Used: CPWD DSR 2023-24 Benchmarks

Calibrated Across: 108 Categories

Real Result: Identified 69,170 cost anomalies representing ₹3,639.9 Crore in excess financial risk.

Feature 4: CHAKRA-VYUH (Contractor Network Cartel Sentry)

NETWORK CARTEL
SENTRY

In Simple Words: A syndicate detector that uncovers vendor cartels, monopoly contracts, and non-competitive bidding networks.

HOW IT WORKS:

- Constructs a full relational network graph indexing **109,521 real MoSPI payment vouchers** and 27,234 registered contractors.
- Calculates vendor concentration ratios per district. If 80%+ of all MP fund disbursements in a constituency flow to a single contractor or linked entity, CHAKRA-VYUH flags a high cartel risk.

Data Used: 109,521 Payment Vouchers

Entities Indexed: 27,234 Contractors & 701 MPs

 **Real Result:** Flagged 47,302 contractor cartel risks across 27,234 commercial vendors nationwide.



Feature 5: VIBHED-NETRA (12-Dimensional Isolation Forest ML Sentry)

MACHINE LEARNING
OUTLIER ENGINE

In Simple Words: An advanced AI pattern analyzer that discovers hidden anomalies that human auditors easily overlook.

HOW IT WORKS:

- Runs an **Isolation Forest Machine Learning Model** analyzing 12 numerical dimensions simultaneously (cost, physical completion %, execution timeframe, vendor history, district averages, sanction month, etc.).
- Instead of looking at 1 rule at a time, VIBHED-NETRA isolates proposals that display multi-dimensional statistical weirdness.

Model: 12-D Isolation Forest

Latency: 12ms High-Speed Inference

 **Real Result:** Detected 40,971 multi-dimensional anomalies (including 10,286 critical statistical outliers).

12
34

Feature 6: SANKHYA-SATYA (Benford's Law & Tender-Splitting Sentry)

FORENSIC MATH &
SMURFING SENTRY

In Simple Words: A mathematical fraud detector that catches "tender-splitting" (smurfing) and artificial budget estimates.

HOW IT WORKS:

- **Tender-Splitting (Smurfing):** Detects proposals pegged intentionally just below the mandatory ₹5 Lakh e-procurement threshold (e.g. ₹4,95,000 or ₹4,98,000) to bypass open competitive e-tendering (**GFR Rule 149**).
- **Benford's Law Forensic Digit Analysis:** Checks the leading digits (1-9) of sanctioned amounts against natural mathematical probability distributions to flag human-fabricated integer estimates.

Forensics: Benford's Law Distribution

Statute: GFR 2017 Rule 149 Thresholds

🚨 **Real Result:** Caught 7,710 tender-splitting threshold evasions and 54,820 round number estimates.



Feature 7: BHU-DRISHTI (ISRO Bhuvan GIS & Ghost Asset Radar)

GEOSPATIAL
SATELLITE RADAR

In Simple Words: A satellite tracking radar that verifies whether projects physically exist on ground maps and aren't "ghost assets".

HOW IT WORKS:

- Plots asset GPS coordinates on **ISRO Bhuvan Geo-Portal** centroids and Esri high-definition satellite imagery.
- **Ghost Asset Radar:** Flags proposals missing mandatory ISRO Bhuvan / eSAKSHI mobile app GPS geotags.
- **Spatial Cluster Radar:** Flags suspicious hyper-local clusters where multiple works are logged within <250 meters of each other.

GIS: ISRO Bhuvan & Esri HD Satellite

Assets Mapped: 176,925 Centroids

🚨 **Real Result:** Mapped 176,925 geocoded assets, flagging 3,031 ghost assets and 12,320 spatial clusters.



Enterprise Core: Simulator, Form GFR-19A Memorandum & Provenance

DECISION-
SUPPORT CORE

In Simple Words: The complete operational workflow that turns AI alerts into legal, actionable executive orders.

KEY TOOLS:

-  **Live Proposal Simulator (<50ms):** An interactive sandbox allowing DMs to input proposal parameters and run instant multi-sentinel simulations *before* sanction approval. Includes 4 one-click demo presets for live pitching.
-  **Form GFR-19A Printable Vigilance Memorandum:** Generates an official printable memorandum complete with MoSPI emblem, barcode, statutory rule citations, 5-point physical engineer verification checklist, and DM signoff block.
-  **Verifiable Data Lineage & Provenance Ledger:** Displays an immutable cryptographic dataset hash (**SHA-256:** 9a5c8df1b038c352...) proving 100% real data integrity under strict ****Human-in-the-Loop (HITL)**** governance.

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Developed with Excellence for Public Financial Integrity & Smart Governance